

Research Funding and Higher Degree by Research Completions

An analysis of Australian higher education providers, 2015–2024

Insight report — Research KPI Intern task

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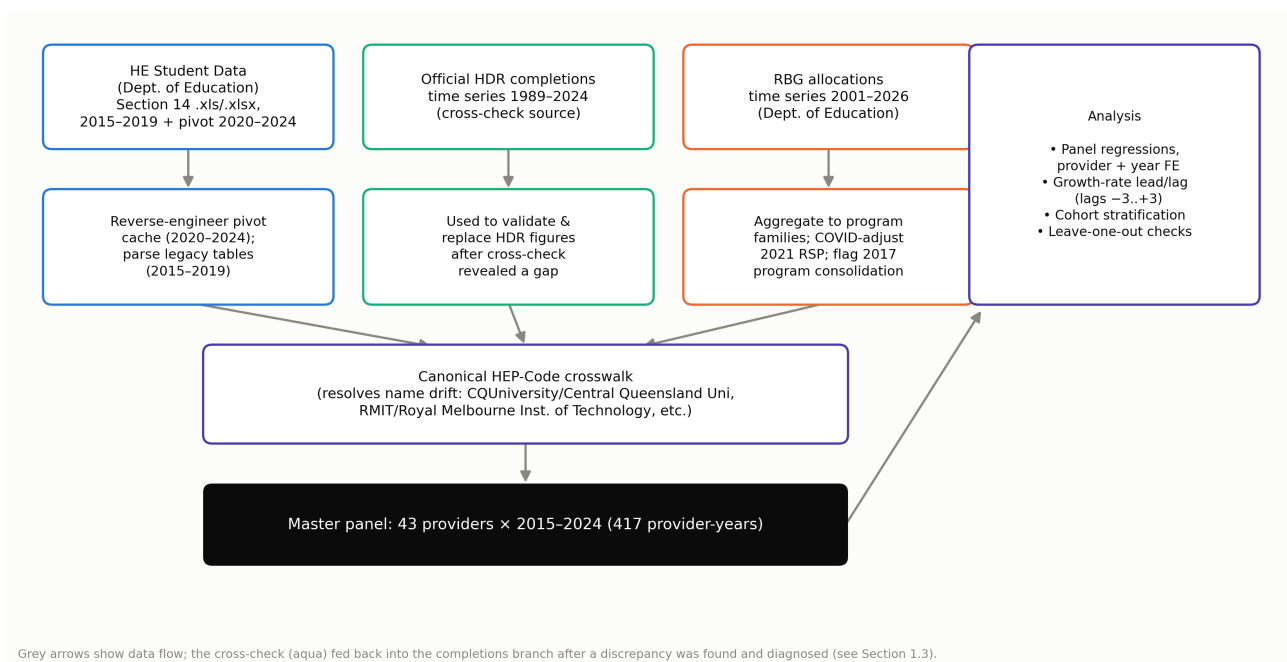
Section 1 — Data and Model

1.1 Data sources

This analysis uses two public data series published by the Australian Government Department of Education, plus one file provided directly for this task as an independent validation source.

- **Higher Education Statistics** (Student Data collection) — <https://www.education.gov.au/higher-education-statistics> — used for Higher Degree by Research (HDR) completions: Doctorate by Research + Master's by Research, by provider and year.
- **Research block grant (RBG) allocations time series** — <https://www.education.gov.au/research-block-grants/resources/research-block-grant-allocations-time-series> — used for research funding: the Research Training Program (RTP) and Research Support Program (RSP), by provider and year, 2001–2026.
- **Official HDR student completions time series (1989–2024)**, supplied directly by the task sponsor, used as an independent cross-check of the completions figures reconstructed from the Student Data publications (Section 1.3).
- **RSP/RTP calculation methodology** and the **R&D income (HERDC) time series**, both published by the Dept. of Education under research-block-grants/resources, used in Section 2 Finding 2 to test whether the funding premium is mechanically explained by the legislated allocation formulas.

1.2 Pipeline



Both raw sources needed non-trivial reconstruction before they could be joined: the RBG file is a clean tidy table, but the 2020–2024 completions figures existed only inside an Excel PivotTable, so the underlying record-level data had to be recovered from the workbook's pivot-cache XML; 2015–2019 completions came from five separate annual publications with inconsistent table numbering.

1.3 Structural issues identified and resolved

- **Provider name drift.** The same institution appears under different strings across years/files (CQUniversity vs. Central Queensland University; RMIT University vs. its RBG legal name Royal Melbourne Institute of Technology; Curtin University vs. Curtin University of Technology). Resolved with a canonical HEP-Code crosswalk anchored on the RBG file's provider list.
- **2017 RBG program consolidation.** Six legacy programs (APA, IPRS, RTS, JRE, RIBG, SRE) were replaced by RTP and RSP in 2017. Resolved by aggregating to these two program families so the series is comparable before and after the reform.

- **2021 COVID-19 anomaly.** RSP allocations roughly doubled in 2021 (a documented one-off \$1bn pandemic support payment) while national HDR completions dipped, plausibly reflecting submission extensions. Both sides of the funding–completions relationship are affected in this single year, so 2021 is excluded from the primary growth-rate specification (included as a robustness check).
- **Cell suppression.** Small counts in the legacy tables are masked ('<5' / 'np'). Handled with documented interval-midpoint imputation, flagged per observation.
- **Independent cross-check.** Comparing the reconstructed completions panel against the official HDR file (same HEP-Code scheme as the RBG file) showed a high overall level correlation ($r=0.99$) but material gaps concentrated in 2021 and a few other institution-years (e.g. Sydney –300, Melbourne –334). Diagnosis: the reconstructed panel's *total* (all-level) completions for these same institution-years matched an independently published all-level series almost exactly, so the crosswalk and overall counts were correct — the gap was specific to the Excel pivot tool's coarse course-level bucketing of HDR completions in COVID backlog-clearing years. The completions side of the panel was rebuilt directly from the official, purpose-built HDR file for the full analysis window; the RBG-side cleaning and crosswalk were unchanged. This cross-check is treated as a validation step for the panel-building methodology, not a limitation of it.

1.4 Analytical approach

The cleaned panel covers **43 providers × 2015–2024 (417 provider-years)**. Relationships are estimated with log-linear OLS carrying provider and year fixed effects and standard errors clustered by provider, both in levels (funding on completions, testing for a cross-sectional concentration premium) and in year-on-year growth rates (testing lead/lag dynamics at lags –3 to +3, both directions and contemporaneously). Findings are stress-tested against: dropping 2021; splitting RTP vs. RSP; adding total (all-level) completions as an alternative scale control; stratifying by mission group (Go8, ATN, IRU, RUN, non-aligned); and a leave-one-Go8-university-out check.

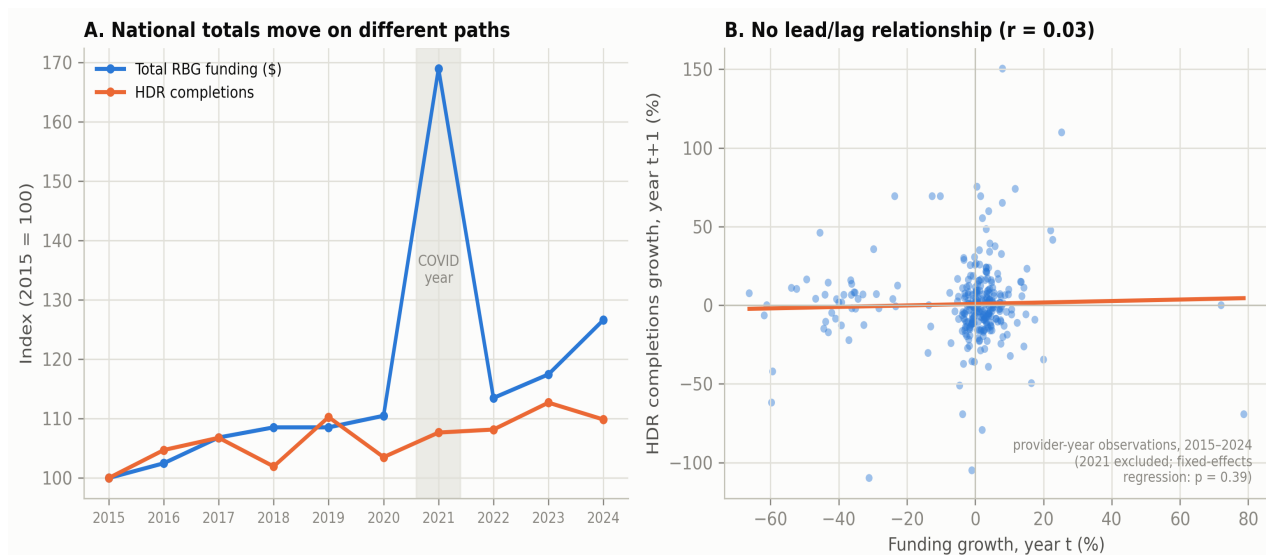
1.5 Limitations

- **Window length.** Ten annual observations per provider is short for detecting slow-moving dynamics; a relationship operating on a 3–5 year cycle (plausible given RTP's own multi-year formula) may not be visible in year-on-year growth rates even if it exists.
- **Correlational, not causal.** Fixed effects remove level differences between providers and common shocks by year, but do not identify causal effects on their own — an unobserved provider-specific trend correlated with both funding and completions growth could still confound either finding.
- **Small-provider suppression.** Cell-count masking in the legacy tables is concentrated among the smallest providers, who contribute little to the funding totals driving either finding, but their individual growth rates are noisier.
- **Forward changes not in scope.** The 2026 merger of the University of Adelaide and University of South Australia into “Adelaide University” falls outside the 2015–2024 window and does not affect either finding, but would need a crosswalk update before extending this panel forward.

Section 2 — Key Findings

FINDING 1

No statistically significant funding–completions dynamic detected — the apparent link is a scale artifact



Source: Dept. of Education RBG allocations time series; HDR completions time series (1989–2024). $n=417$ provider-years, 2015–2024 ($n=850$, 2001–2024 for the extended-window check).

Raw provider-level correlation between RBG funding and HDR completions is high ($r \approx 0.91$) — the kind of number that invites a quick “funding drives completions” headline. It doesn’t survive scrutiny. That correlation is measured in levels, where larger universities mechanically have more of everything: more funding, more completions, more staff, more students. It says nothing about whether a change in one precedes a change in the other.

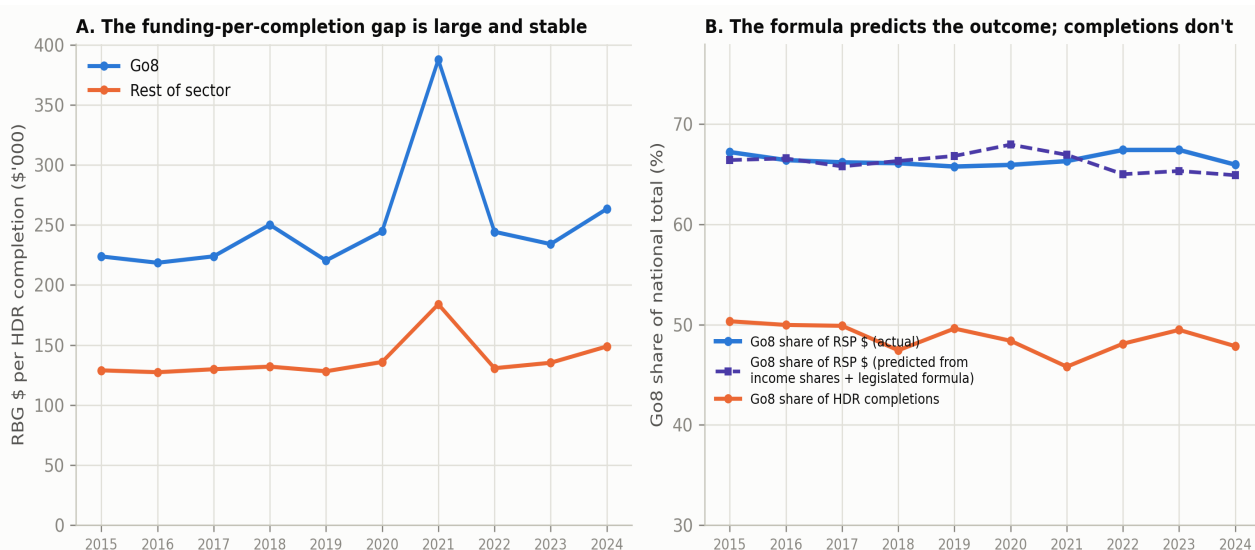
Testing the actual dynamic — does year-on-year funding growth lead, lag, or move with completions growth — removes the scale effect via provider and year fixed effects and tests lags -3 to $+3$ in both directions, plus the same-year relationship. Essentially none of it is significant: funding growth $\rightarrow$ next-year completions growth (coef= -0.33 , $p=0.39$); completions growth $\rightarrow$ next-year funding growth (coef= ≈ 0.00 , $p=0.96$); same-year ($p=0.14$). This holds with Total RBG or RTP alone, with/without 2021, and — tested after this draft was first reviewed — holds even more strongly on an extended 2001–2024 window ($n=850$ vs. 290 , $p=0.29$ – 0.71 across variants). One isolated exception: the same-year relationship turns marginally significant ($p=0.03$, a negative coefficient) specifically when 2024 is excluded — not replicated in the lead specification or the extended window, and consistent with chance given the number of specifications tested.

Panel A shows why: indexed to 2015, RBG funding climbs (and spikes sharply in the 2021 COVID top-up) while HDR completions grows on a materially different, choppy path — even a 69% one-year funding jump produced no matching movement in completions. Panel B shows the same result at the provider level: funding growth and next-year completions growth form a flat, uncorrelated cloud ($r=0.03$).

So what: RTP’s own allocation formula is explicitly a multi-year rolling average, not a same-year or one-year-lag mechanism — this result is consistent with that design, not a contradiction of it, but it means single-year funding movements are not a usable signal for forecasting near-term completions (or vice versa) at the provider level. Anyone building a KPI dashboard around annual funding-to-output tracking should not expect it to move in lockstep.

FINDING 2

Go8's funding premium per completion is not about completions — it's the legislated formula's income weighting



Source: Dept. of Education RBG allocations time series; HDR completions time series (1989–2024); R&D income (HERDC) time series; RSP/RTP calculation guidance. $n=417$ provider-years, 2015–2024.

Group of Eight (Go8) universities — 8 of the 43 providers in the panel — hold roughly 48% of the sector's HDR completions but 62–64% of RBG dollars, a gap that has held steady across all ten years and appears independently at every single Go8 university (1.56x–2.24x the non-Go8 \$-per-completion average; not 2–3 outliers carrying the result).

This is mechanical, not incidental. The Department's own legislated formulas (Other Grants Guidelines (Research) 2017; Higher Education Support (Commonwealth Scholarships) Guidelines 2025) show **RSP is 47% competitive R&D income share + 53% engagement income share — with zero HDR completions weighting**, while RTP is 25% + 25% income plus **50% weighted HDR completions**. Go8 holds ~68.5% of national competitive income and ~64.2% of engagement income (2015–2024 avg.) — both far above its ~48% completions share. Reconstructing Go8's predicted RSP share from nothing but these income shares and the legislated weights reproduces its *actual* observed RSP share to within 1–2 percentage points every year (Panel B) — the funding pattern is the formula working as designed, not a completions story at all.

Consistent with this, the regression premium is **+124% in RSP** ($p<0.0001$) but only **+21% in RTP** ($p=0.10$, n.s.) once HDR completions volume is controlled for — precisely because RTP's formula is half-diluted by a completions component RSP doesn't have. (Using total, not HDR-specific, completions as the control flips RTP significant too — the choice of denominator matters, and HDR completions is the one RTP's own formula actually uses.) Other mission groups (ATN, IRU, RUN) show no significant premium over non-aligned providers.

So what: the popular reading — that Go8 dominates because it “out-produces” the sector on research training — gets this backwards. Go8's advantage is concentrated in R&D income (competitive grants and industry engagement), a materially different policy lever than HDR training capacity. A funding-equity intervention aimed at completions would be aiming at the wrong mechanism.

Appendix

The full data pipeline, cleaning scripts, regression code, and chart-generation code for this report are published in a documented Git repository, including the reconciliation crosswalk, the pivot-cache reconstruction, and every robustness check referenced in Section 1.4. The repository also includes a three-page Power BI dashboard extending both findings with interactive slicers and provider-level detail (screenshots in the repository's README; no live web link, as this account's Power BI tenant has "Publish to web" disabled).

Git repository: <https://github.com/mithizs08/UTS-Research-KPI-Analysis>

Note on reproducibility: all figures in this report are computed directly from the two published Department of Education time series plus the official HDR completions file supplied for this task; no manual data entry was used. Suppressed-cell imputation and the 2021 COVID exclusion are documented choices, not silent adjustments — both are described in Section 1.3 and implemented as explicit, labelled steps in the repository code.